Week 7: Data workflows need a risk register before any robot

Weekly evidence artifact · Week 7 · Trimester 1, 2026

1. Theory anchor

Data governance and risk register [10].

2. Artifact

Risk	Control	Owner
Sensor drift or failure	Revert to manual rounds at 3% drift	Lead Engineer
TITR API outage	Manual customs filing	Logistics Officer
Robot fault at Ulba	Shielded manual tele-manipulator	Plant Safety Manager

Workflow: sensor → filter → store → model → alert → human review → action log.

3. Company A vs. Company B insight

Cameco uses remote robotics in dangerous areas [3]. Only the drum-packing idea transfers to Ulba.

4. Transfer verdict

Pilot one robotic drum-packing arm at Ulba; **Reject** underground jet-boring robots.

5. Evidence + verification

Robotics source [3]. Target: 30% fewer worker radiation hours over 6 months, a **proposed pilot target**.

6. Mini-recommendation

The Ulba Plant Safety Manager should own the robot pilot; progress signal: monthly exposure hours.

7. Learning progression & AI disclosure

I now see governance as part of the design, not a final step. AI drafted the risk-table format; I chose the risks and controls.

Companies: Cameco (A) and NAC Kazatomprom JSC (B). Source numbers [n] refer to the Evidence page of the ePortfolio. Proposed targets and thresholds are design assumptions and pilot targets, not verified Kazatomprom data.